Microsoft

Microsoft Build

May 7–9, 2018 // Seattle, WA

Microsoft

What's New in Xamarin.Forms 3.0

David Ortinau
Senior Program Manager
Xamarin.Forms

BRK2437

2017 Quick Recap

2017 Quick Recap

Public Roadmap

aka.ms/xfroadmap

GitHub

github.com/xamarin/Xamarin.Forms

Issues

Projects

33 Releases

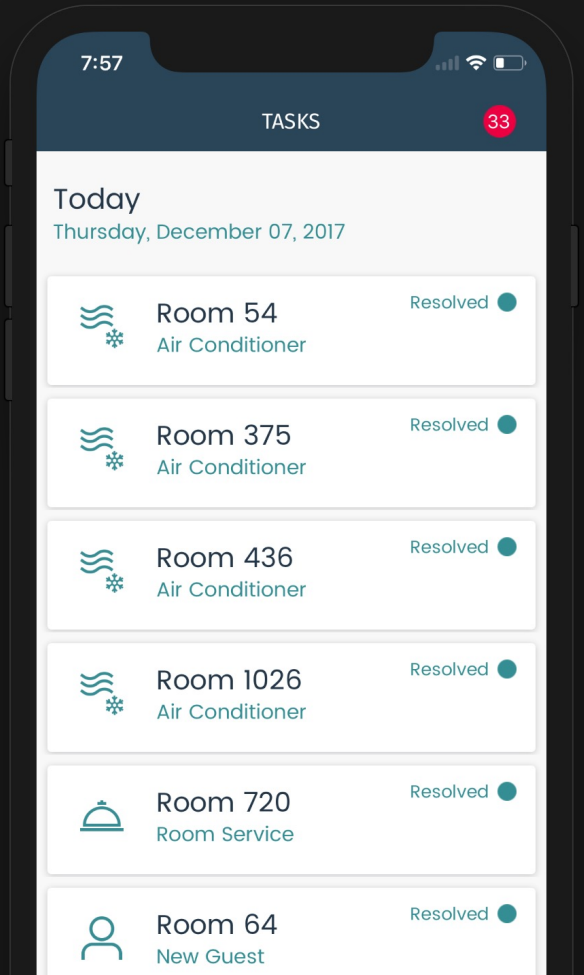
16 Pre-Releases

17 Stable Releases

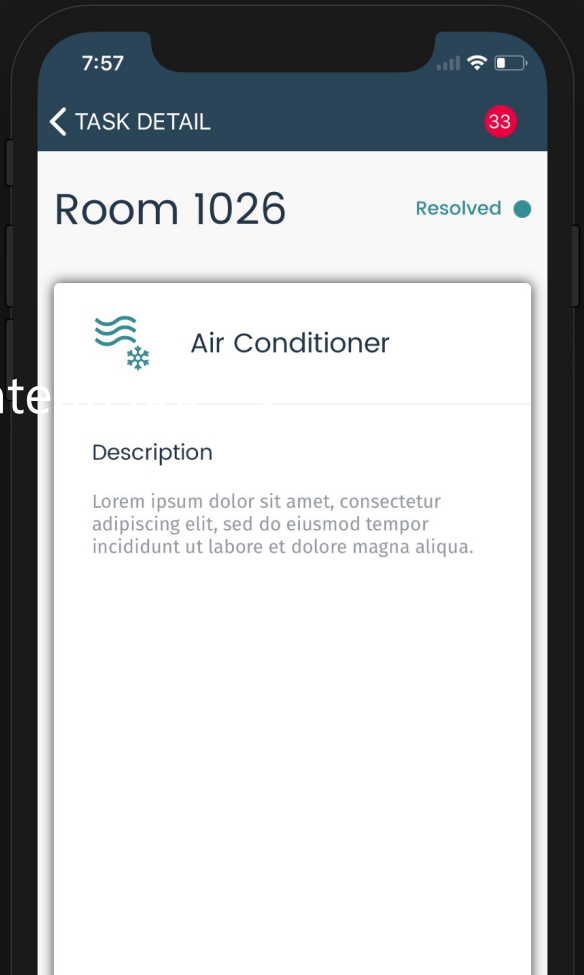
Native Forms

Forms UI embedded in Xamarin.iOS, Xamarin.Android, UWP, etc.

UIViewController ->

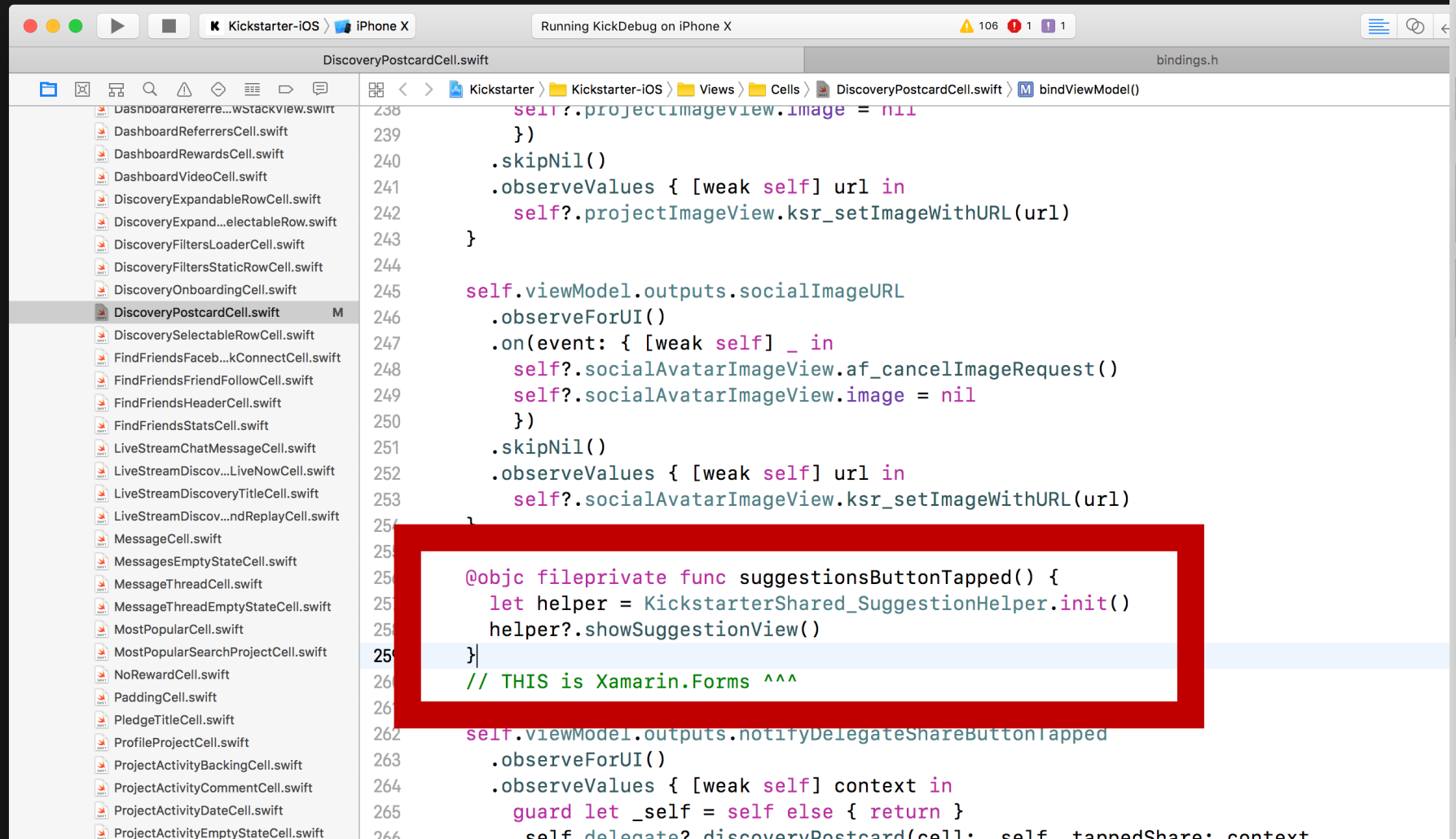


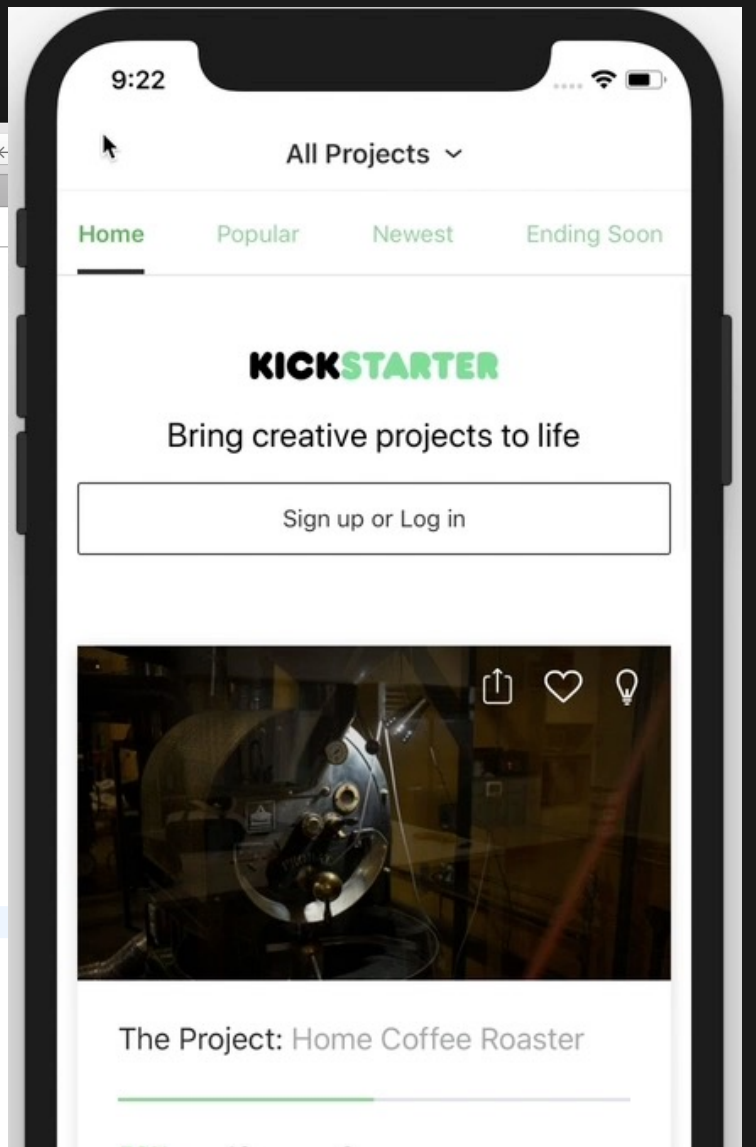
-> Conte



Native Forms + .NET Embedding

Forms UI and .NET embedded in Obj-C





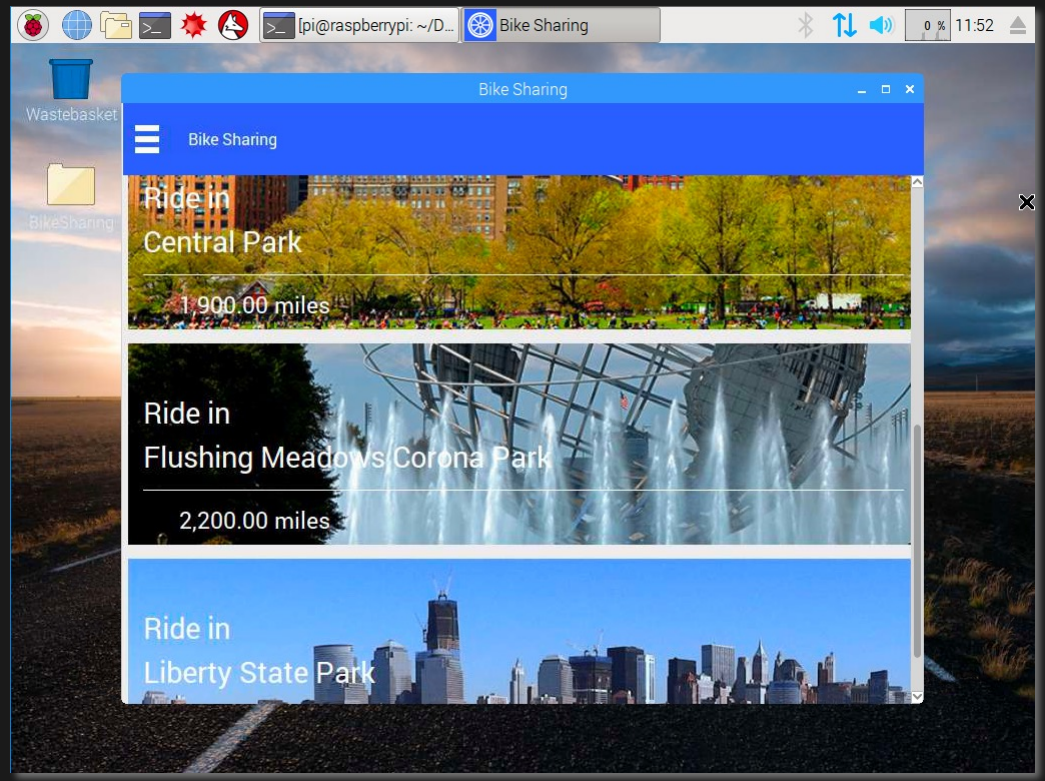
Performance

- .NET Standard. 2.0
- Compiled Bindings
- Fast Renderers
- Layout Compression
- Startup Optimizations
- XAMLC

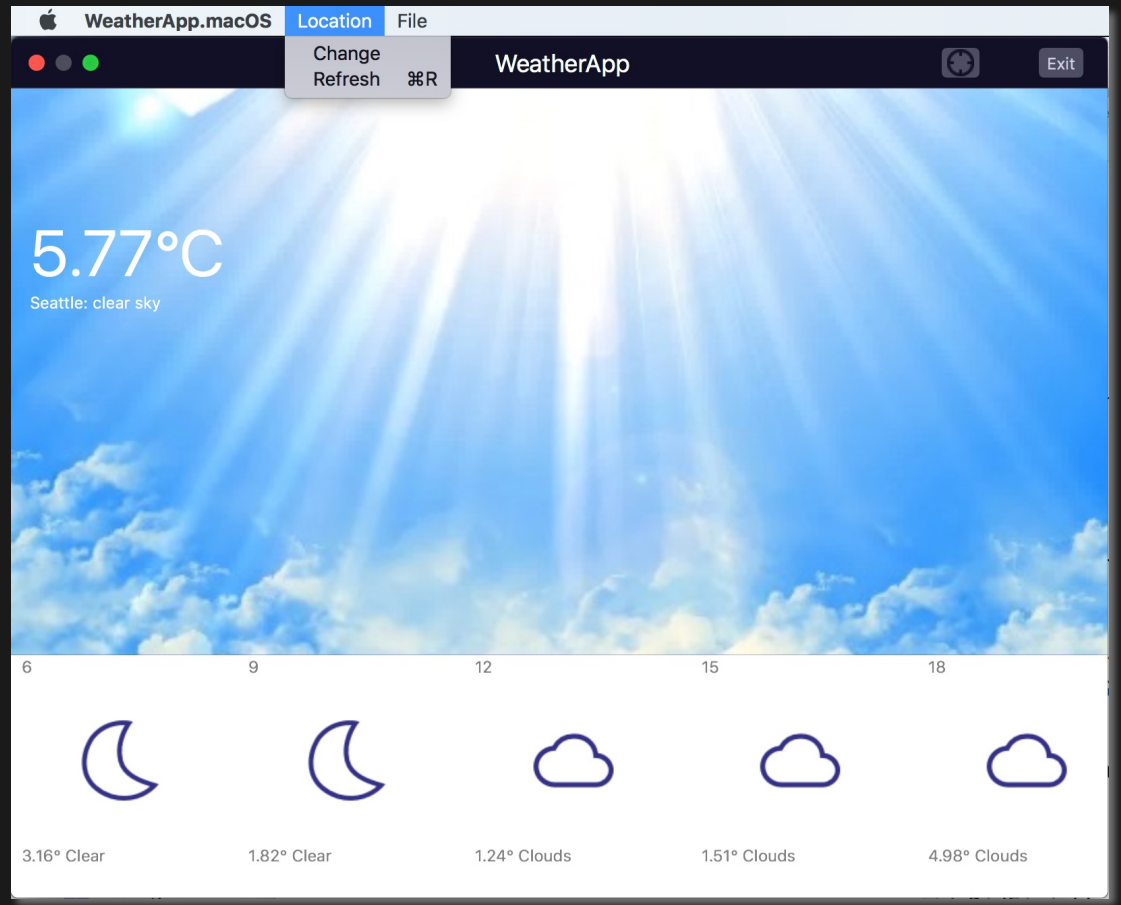
Community Contributions

Contributions FTW!

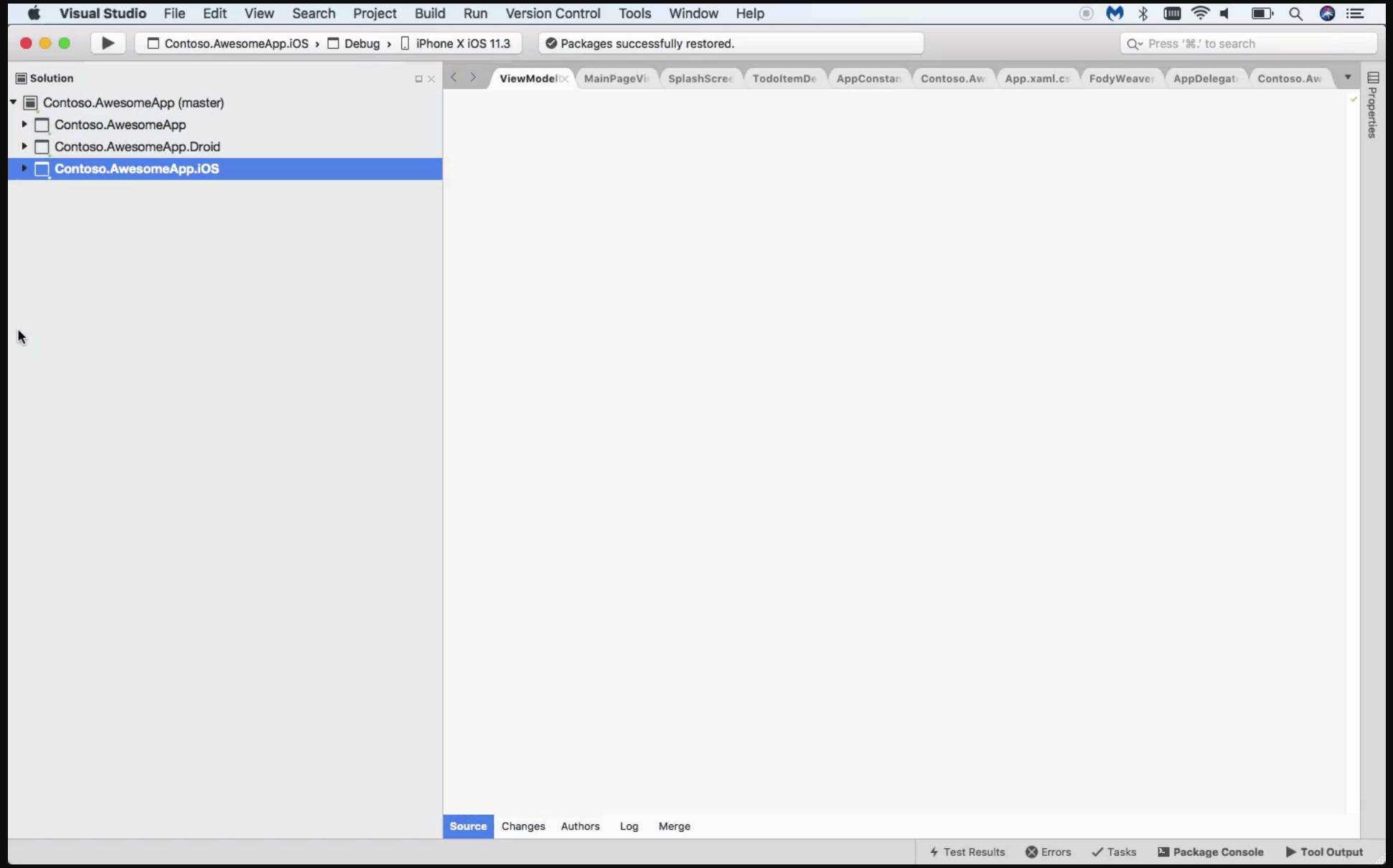
133 Contributors

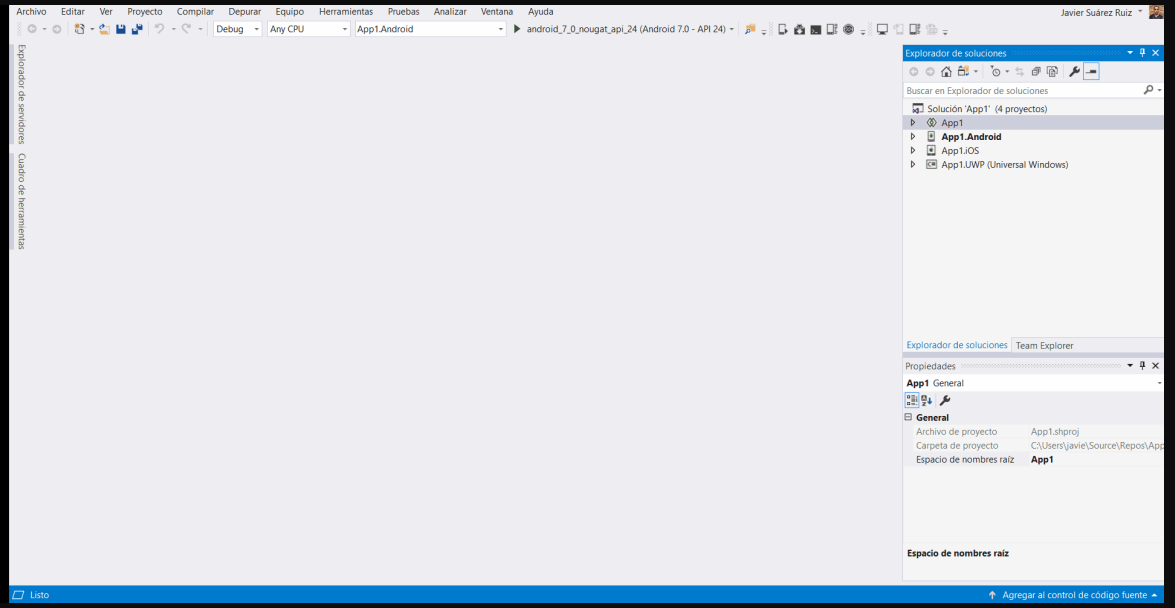


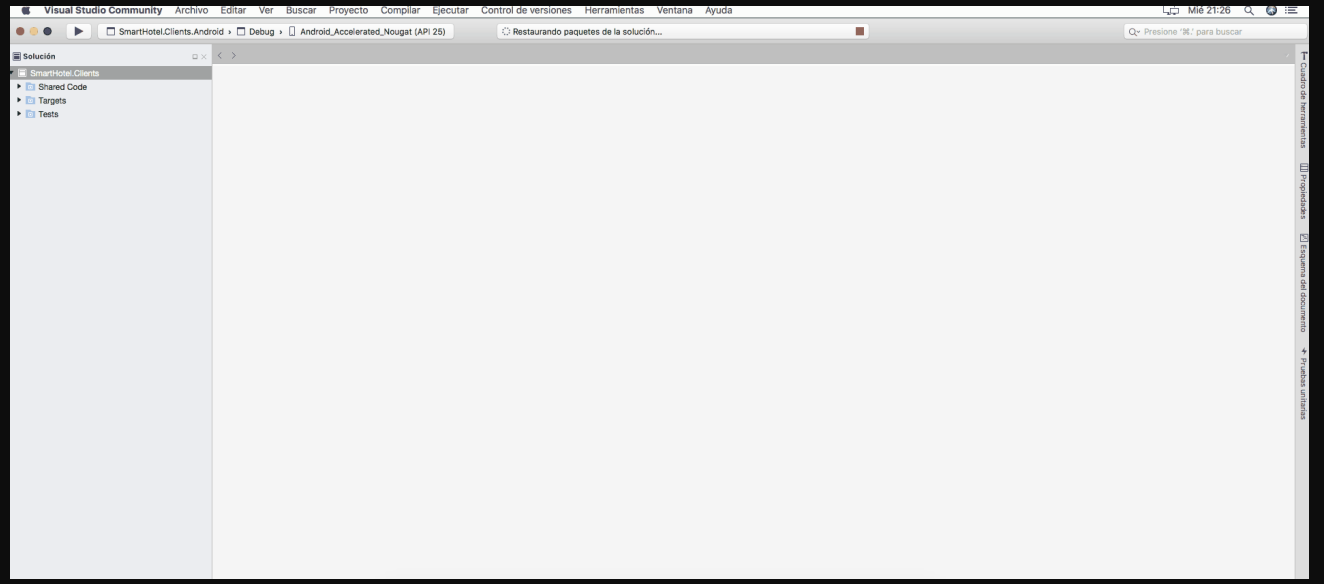
GTK#



macOS


Visual Studio File Edit View Search Project Build Run Version Control Tools Window Help
Contoso.AwesomeApp.iOS › Debug › iPhone X iOS 11.3
Packages successfully restored.
Press '⌘.' to search
Solution
Contoso.AwesomeApp (master)
Contoso.AwesomeApp
Contoso.AwesomeApp.Droid
Contoso.AwesomeApp.iOS
ViewModel
MainPageVi
SplashScre
TodoItemDe
AppConstan
Contoso.Aw
App.xaml.cs
FodyWeaver
AppDelegate
Contoso.Aw
Properties
Source
Changes
Authors
Log
Merge
Test Results
Errors
Tasks
Package Console
Tool Output


Archivo Editar Ver Proyecto Compilar Depurar Equipo Herramientas Pruebas Analizar Ventana Ayuda
Javier Suárez Ruiz
Debug
Any CPU
App1.Android
android_7_0_nougat_api_24 (Android 7.0 - API 24)
Explorador de servidores
Cuadro de herramientas
Explorador de soluciones
Buscar en Explorador de soluciones
Solución 'App1' (4 proyectos)
App1
App1.Android
App1.iOS
App1.UWP (Universal Windows)
Explorador de soluciones
Team Explorer
Propiedades
App1 General
General
Archivo de proyecto
App1.shproj
Carpeta de proyecto
Espacio de nombres raíz
App1
Espacio de nombres raíz
Listo
Agregar al control de código fuente


Visual Studio Community
Archivo Editar Ver Buscar Proyecto Compilar Ejecutar Control de versiones Herramientas Ventana Ayuda
Mié 21:26
SmartHotel.Clients.Android
Debug
Android_Accelerated_Nougat (API 25)
Restaurando paquetes de la solución...
Presione '⌘;' para buscar
Solución
SmartHotel.Clients
Shared Code
Targets
Tests
Cuadro de herramientas
Propiedades
Explorador de soluciones
Notificaciones

Platform Support

Edit New Page

David Ortinau edited this page 9 days ago · 4 revisions

Successful open source requires the strong participation of several parties. We want to make clear who is expected to maintain the different pieces of this project, and track the current status of each Xamarin.Forms platform backend.

Repository Maintainer: Microsoft Xamarin.Forms Team
Program Manager: David Ortinau
Engineering Team Lead: Jason Smith

Microsoft fully supports Xamarin.Forms core features which includes data binding, navigation, layouts, control abstractions, and services.

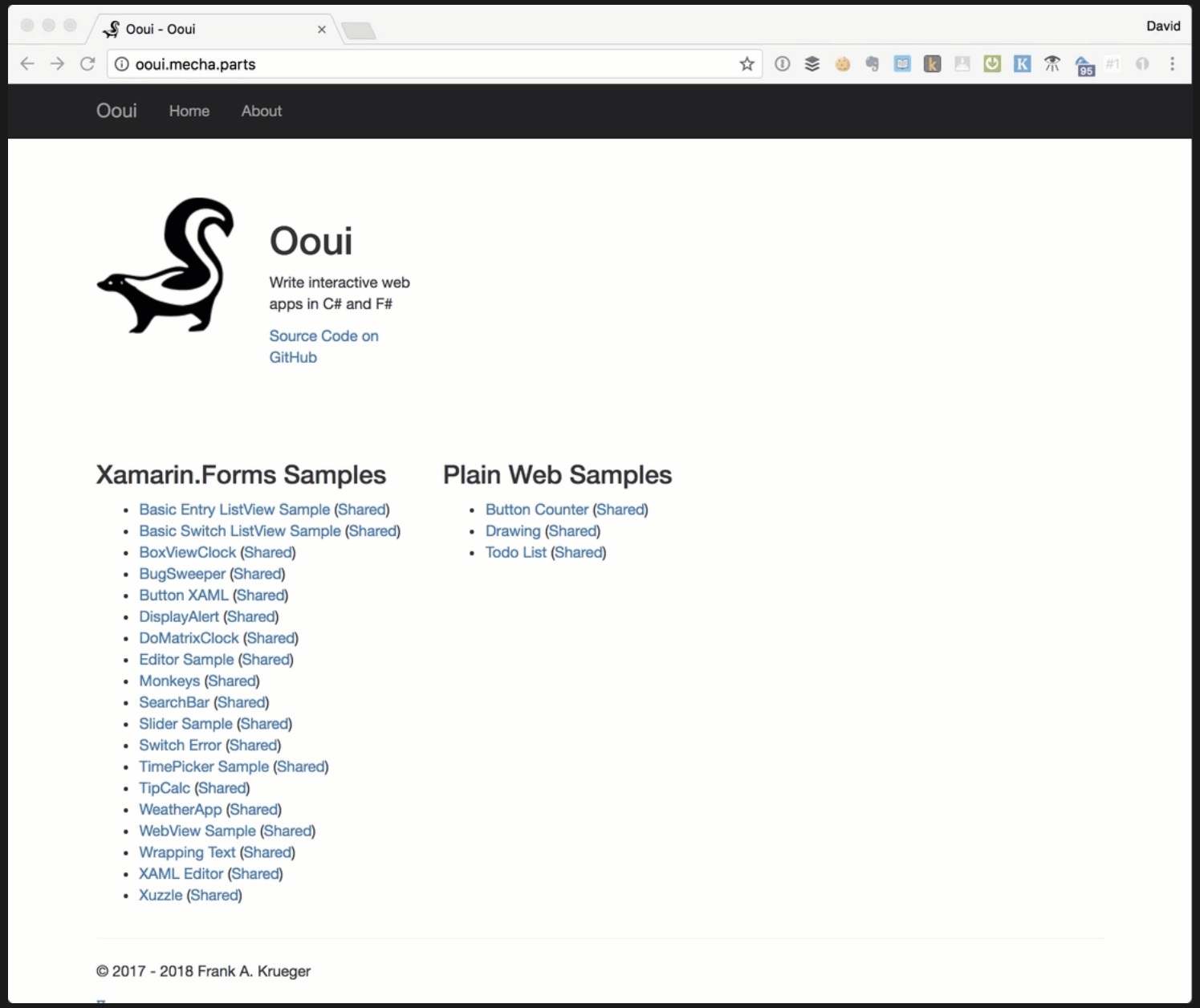
Platform Backend	Status	Supported By
Android	Stable	Microsoft
GTK#	Early Preview - Report	Community
iOS	Stable	Microsoft
macOS	Preview	Microsoft
Tizen	Stable	Third Party - Samsung
UWP	Stable	Microsoft
WPF	Early Preview - Report	Community

Status

Early Preview - these platforms are early in the development cycle and made available for

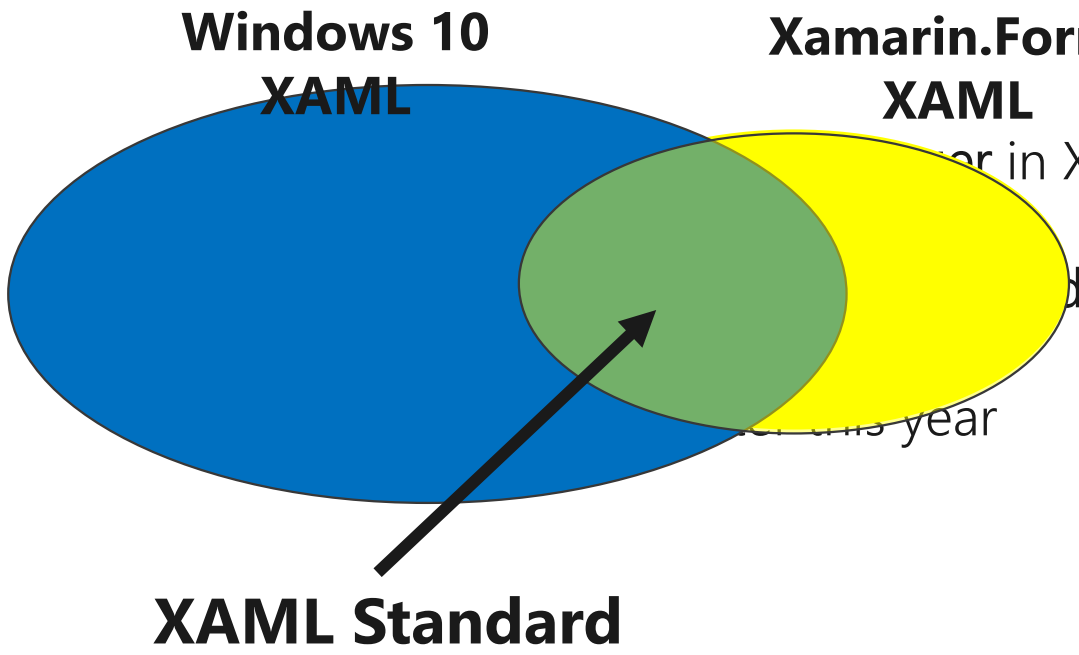
Pages 7
Home
Feature Roadmap
How to Build Xamarin.Forms NuGet
How to Contribute
Nightly Builds
Platform Support
Submitting Issues
Add a custom sidebar

Clone this wiki locally
https://github.com/xamar:
Clone in Desktop


Ooui - Ooui
David
ooui.mecha.parts
Ooui
Home
About
Ooui
Write interactive web apps in C# and F#
Source Code on GitHub
Xamarin.Forms Samples
Basic Entry ListView Sample (Shared)
Basic Switch ListView Sample (Shared)
BoxViewClock (Shared)
BugSweeper (Shared)
Button XAML (Shared)
DisplayAlert (Shared)
DoMatrixClock (Shared)
Editor Sample (Shared)
Monkeys (Shared)
SearchBar (Shared)
Slider Sample (Shared)
Switch Error (Shared)
TimePicker Sample (Shared)
TipCalc (Shared)
WeatherApp (Shared)
WebView Sample (Shared)
Wrapping Text (Shared)
XAML Editor (Shared)
Xuzzle (Shared)
Plain Web Samples
Button Counter (Shared)
Drawing (Shared)
Todo List (Shared)
© 2017 - 2018 Frank A. Krueger

XAML

XAML Standard Update



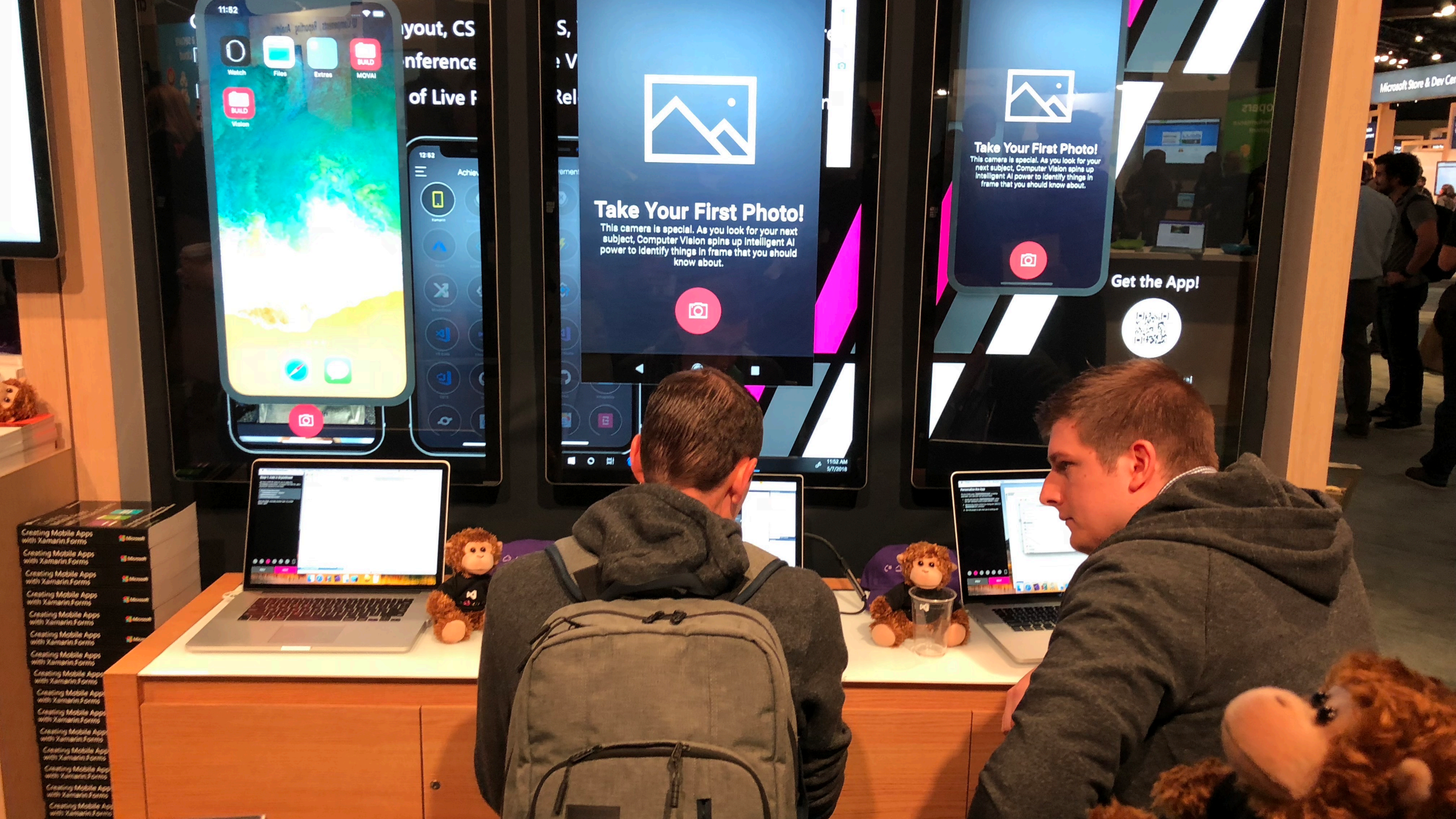
in Xamarin.Forms 3.0

tags with Xamarin.Forms 3.1

year

Xamarin.Forms 3.0


Xamarin.Form
Go hands-on with FlexLayout, CS
Personalize your own Conferenc
Preview the productivity of Live
s 3.0
S, VisualStateManager, and more!
Vision app in 5 minutes.
Reload as you design your screens.
Get the App!
Microsoft
Applications


Take Your First Photo!
This camera is special. As you look for your next subject, Computer Vision spins up intelligent AI power to identify things in frame that you should know about.
Take Your First Photo!
This camera is special. As you look for your next subject, Computer Vision spins up intelligent AI power to identify things in frame that you should know about.
Get the App!

Conference Vision Demo

Get the App!



https://aka.ms/movai

Main Features

Visual State Manager

Right-To-Left

ResourceDictionary Improvements

FlexLayout

CSS

F100

Visual State Manager

Visual State Manager

- VisualElement “CommonStates”
  - Normal
  - Focused
  - Disabled

- GoToState(this, “Focused”);

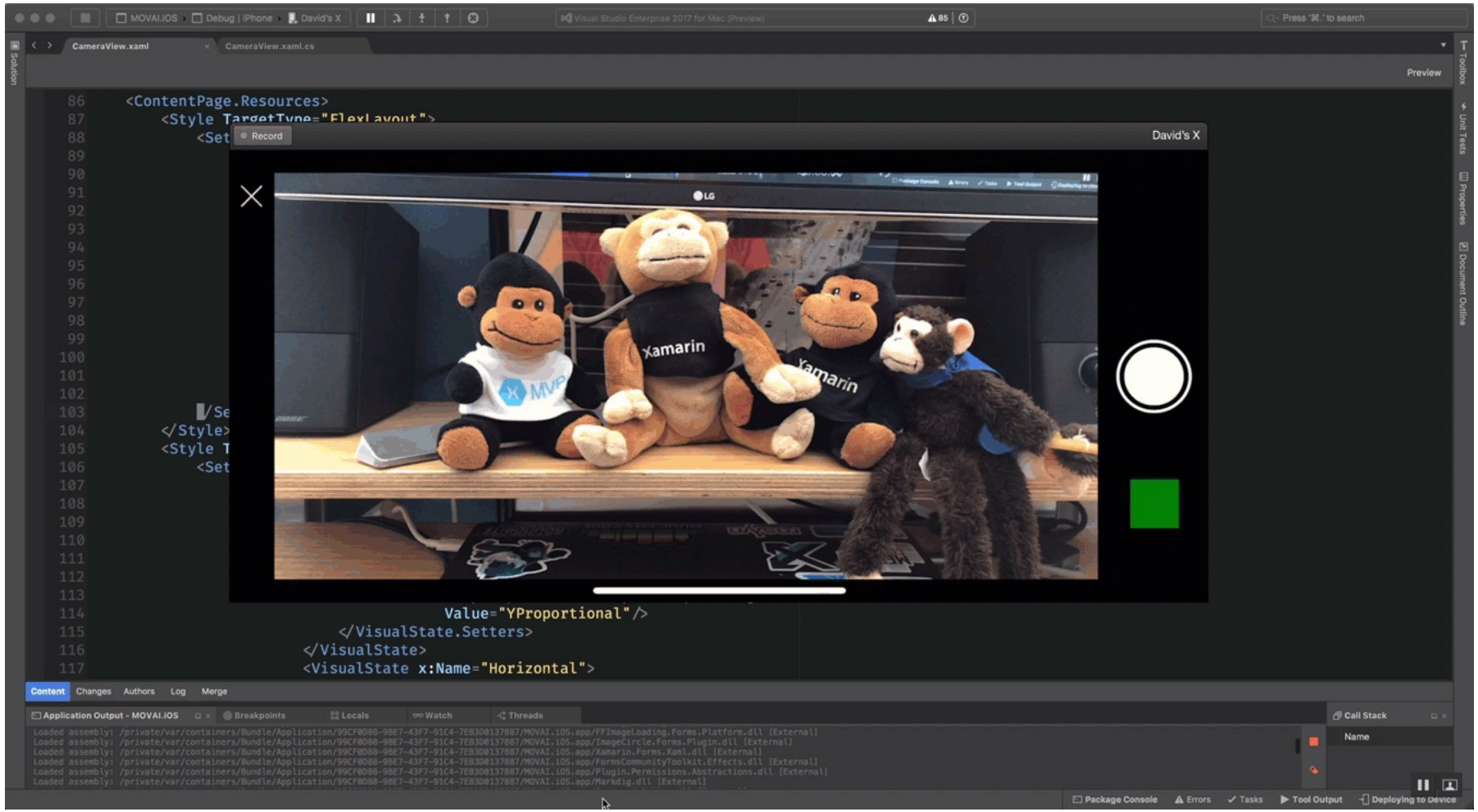
```
<Entry FontSize="18">
    <VisualStateManager.VisualStateGroups>
        <VisualStateGroup x:Name="CommonStates">
            <VisualState x:Name="Normal">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Lime" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Focused">
                <VisualState.Setters>
                    <Setter Property="FontSize"
                            Value="36" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Disabled">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Pink" />
                </VisualState.Setters>
            </VisualState>
        </VisualStateGroup>
    </VisualStateManager.VisualStateGroups>
</Entry>
```

Visual State Manager

- VisualElement “CommonStates”
  - Normal
  - Focused
  - Disabled

- GoToState(this, “Focused”);

```
<Entry FontSize="18">
    <VisualStateManager.VisualStateGroups>
        <VisualStateGroup x:Name="CommonStates">
            <VisualState x:Name="Normal">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Lime" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Focused">
                <VisualState.Setters>
                    <Setter Property="FontSize"
                            Value="36" />
                </VisualState.Setters>
            </VisualState>
            <VisualState x:Name="Disabled">
                <VisualState.Setters>
                    <Setter Property="BackgroundColor"
                            Value="Pink" />
                </VisualState.Setters>
            </VisualState>
        </VisualStateGroup>
    </VisualStateManager.VisualStateGroups>
</Entry>
```


MOVAI.iOS
Debug | iPhone
David's X
Visual Studio Enterprise 2017 for Mac (Preview)
Press '⌘.' to search
CameraView.xaml
CameraView.xaml.cs
Preview
Record
David's X
Xamarin
Xamarin
MVP
<ContentPage.Resources>
<Style TargetType="FlexLayout">
Value="YProportional" />
</VisualState.Setters>
</VisualState>
<VisualState x:Name="Horizontal">
Content
Changes
Authors
Log
Merge
Application Output - MOVAI.iOS
Breakpoints
Locals
Watch
Threads
Call Stack
Name
Package Console
Errors
Tasks
Tool Output
Deploying to Device

```
<ContentPage.Resources>
        <Style TargetType="FlexLayout">

        </Style>
    </ContentPage.Resources>
```

```
<ContentPage.Resources>
        <Style TargetType="FlexLayout">
            <Setter Property="VisualStateManager.VisualStateGroups">
                <VisualStateGroupList x:Name="OrientationStates">









                </VisualStateGroupList>
            </Setter>
        </Style>
    </ContentPage.Resources>
```

```
<ContentPage.Resources>
        <Style TargetType="FlexLayout">
            <Setter Property="VisualStateManager.VisualStateGroups">
                <VisualStateGroupList x:Name="OrientationStates">
                    <VisualStateGroup>
                        <VisualState x:Name="Portrait">



                        </VisualState>
                        <VisualState x:Name="Landscape">



                        </VisualState>
                    </VisualStateGroup>
                </VisualStateGroupList>
            </Setter>
        </Style>
    </ContentPage.Resources>
```

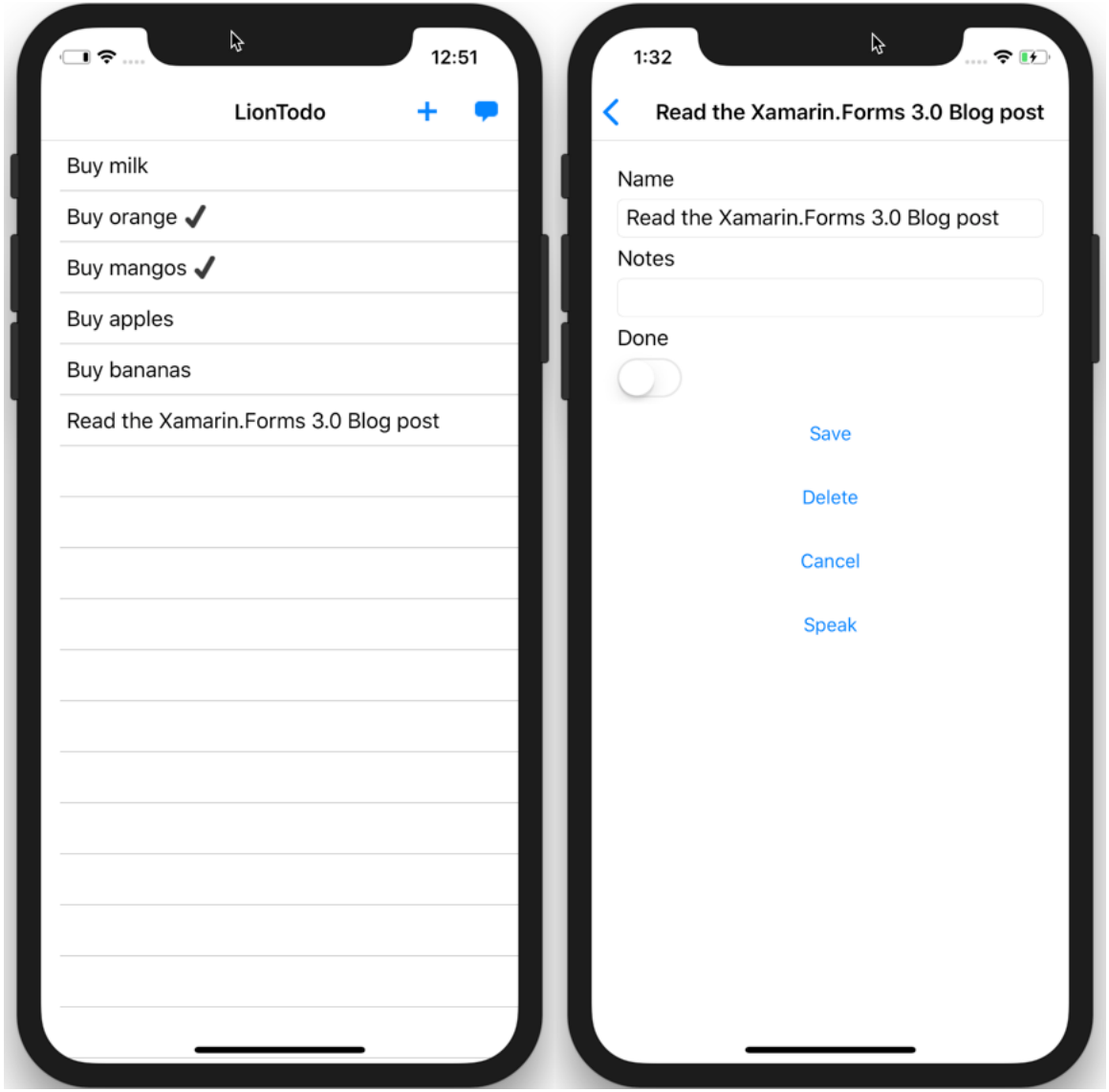
```
<ContentPage.Resources>
        <Style TargetType="FlexLayout">
            <Setter Property="VisualStateManager.VisualStateGroups">
                <VisualStateGroupList x:Name="OrientationStates">
                    <VisualStateGroup>
                        <VisualState x:Name="Portrait">
                            <VisualState.Setters>
                                <Setter Property="Direction" Value="Column”/>
                            </VisualState.Setters>
                        </VisualState>
                        <VisualState x:Name="Landscape">
                            <VisualState.Setters>
                                <Setter Property="Direction" Value="Row”/>
                            </VisualState.Setters>
                        </VisualState>
                    </VisualStateGroup>
                </VisualStateGroupList>
            </Setter>
        </Style>
    </ContentPage.Resources>
```

```
protected override void OnSizeAllocated(double width, double height)
{
    base.OnSizeAllocated(width, height); //must be called
    if (this.width != width || this.height != height)
    {
        this.width = width;
        this.height = height;

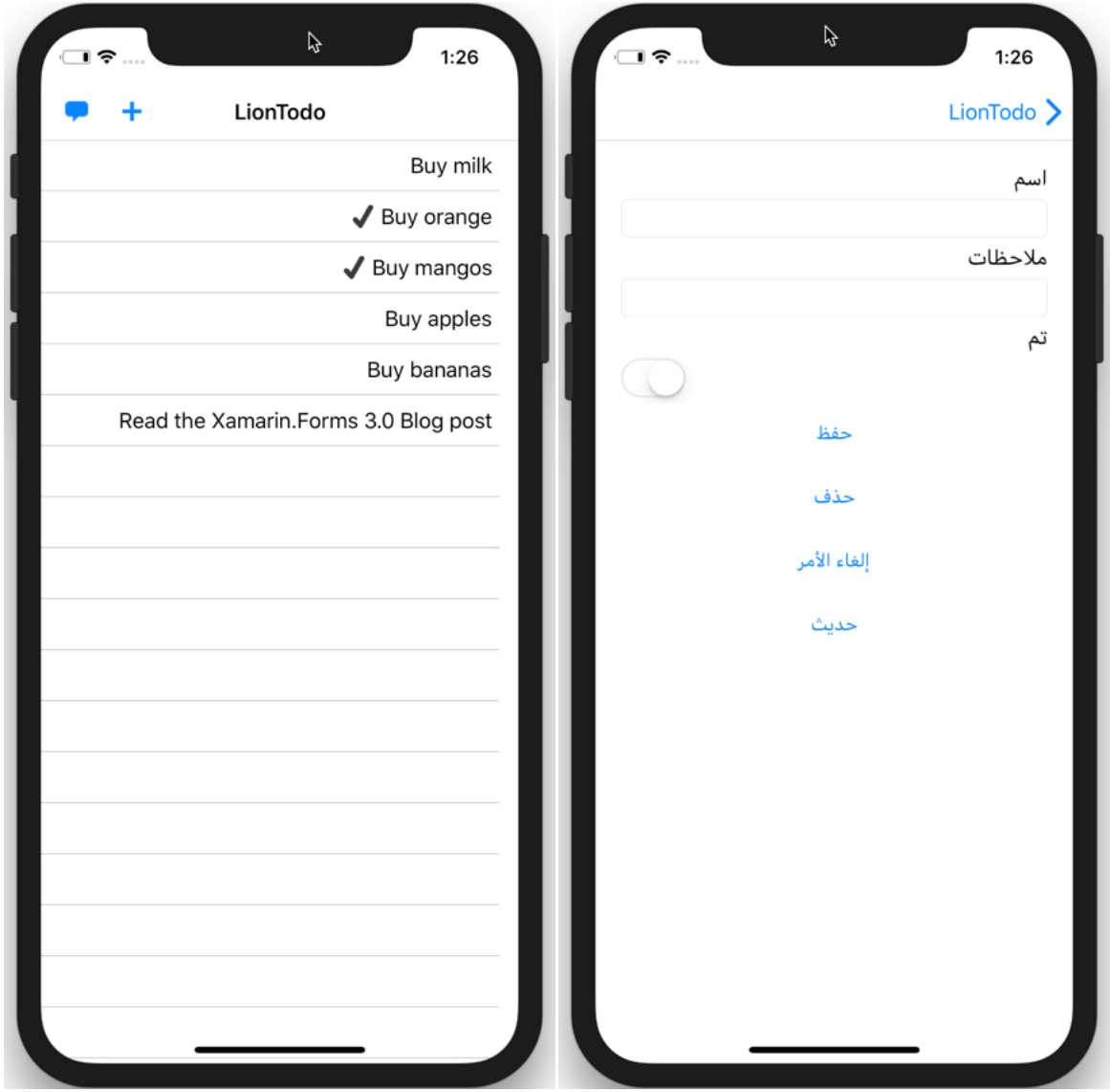
        VisualStateManager.GoToState(Container,
            (width > height) ? "Horizontal" : "Portrait");
        VisualStateManager.GoToState(LastImage,
            (width > height) ? "Horizontal" : "Portrait");
    }
}
```

Right-To-Left

LTR


12:51
LionTodo
Buy milk
Buy orange ✓
Buy mangos ✓
Buy apples
Buy bananas
Read the Xamarin.Forms 3.0 Blog post
1:32
Read the Xamarin.Forms 3.0 Blog post
Name
Read the Xamarin.Forms 3.0 Blog post
Notes
Done
Save
Delete
Cancel
Speak

RTL


1:26
LionTodo
Buy milk
✓ Buy orange
✓ Buy mangos
Buy apples
Buy bananas
Read the Xamarin.Forms 3.0 Blog post
1:26
LionTodo
اسم
ملاحظات
تم
حفظ
حذف
إلغاء الأمر
حديث

FlowDirection="{x:Static Device.FlowDirection}"

FlexLayout + CSS

FlexLayout

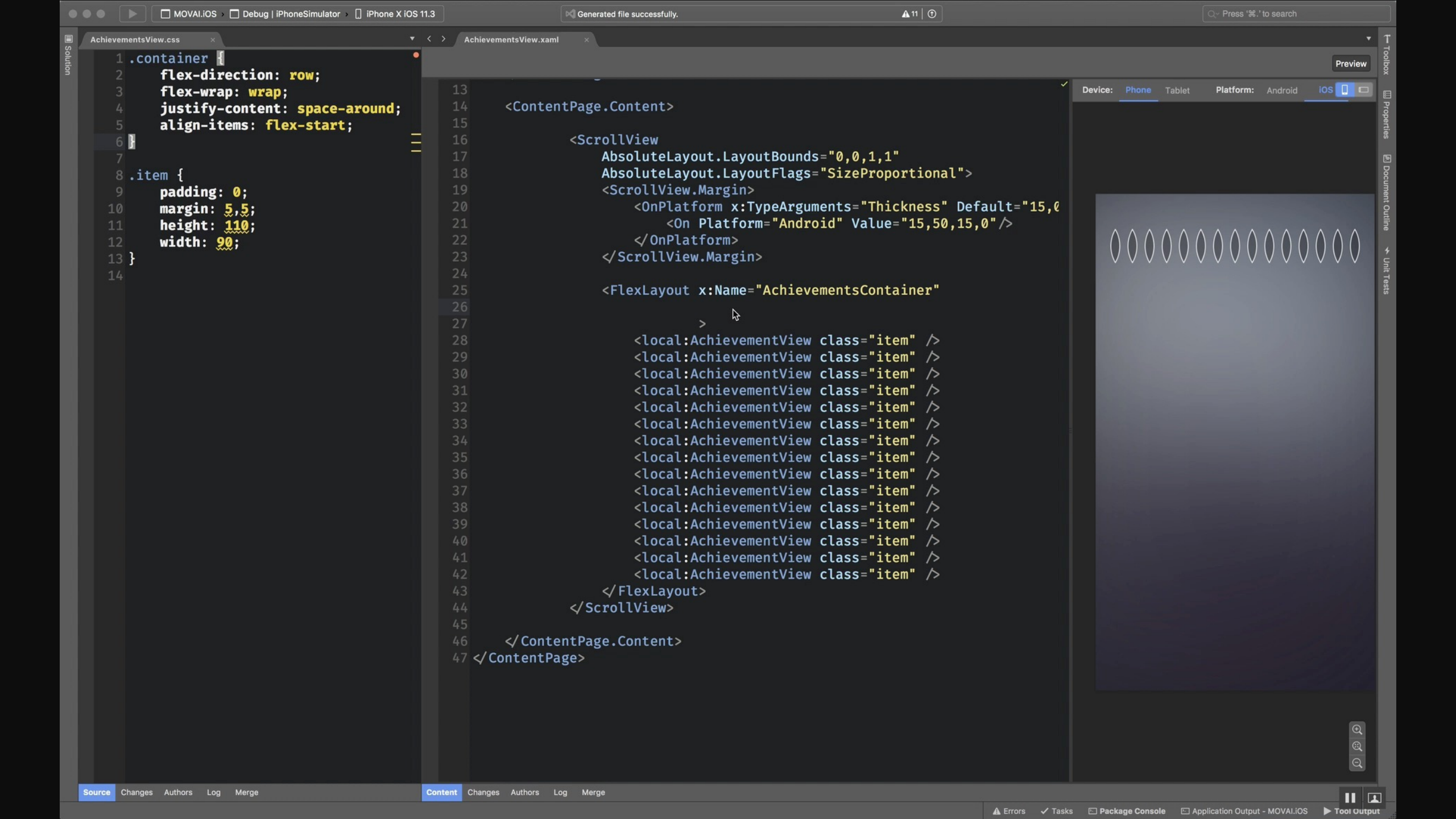
- Flexbox inspired layout system
- Prioritizes distribution and alignment
- Great for adaptive layouts – multiple form factors
- Promotes flatter layout, reducing UI nesting

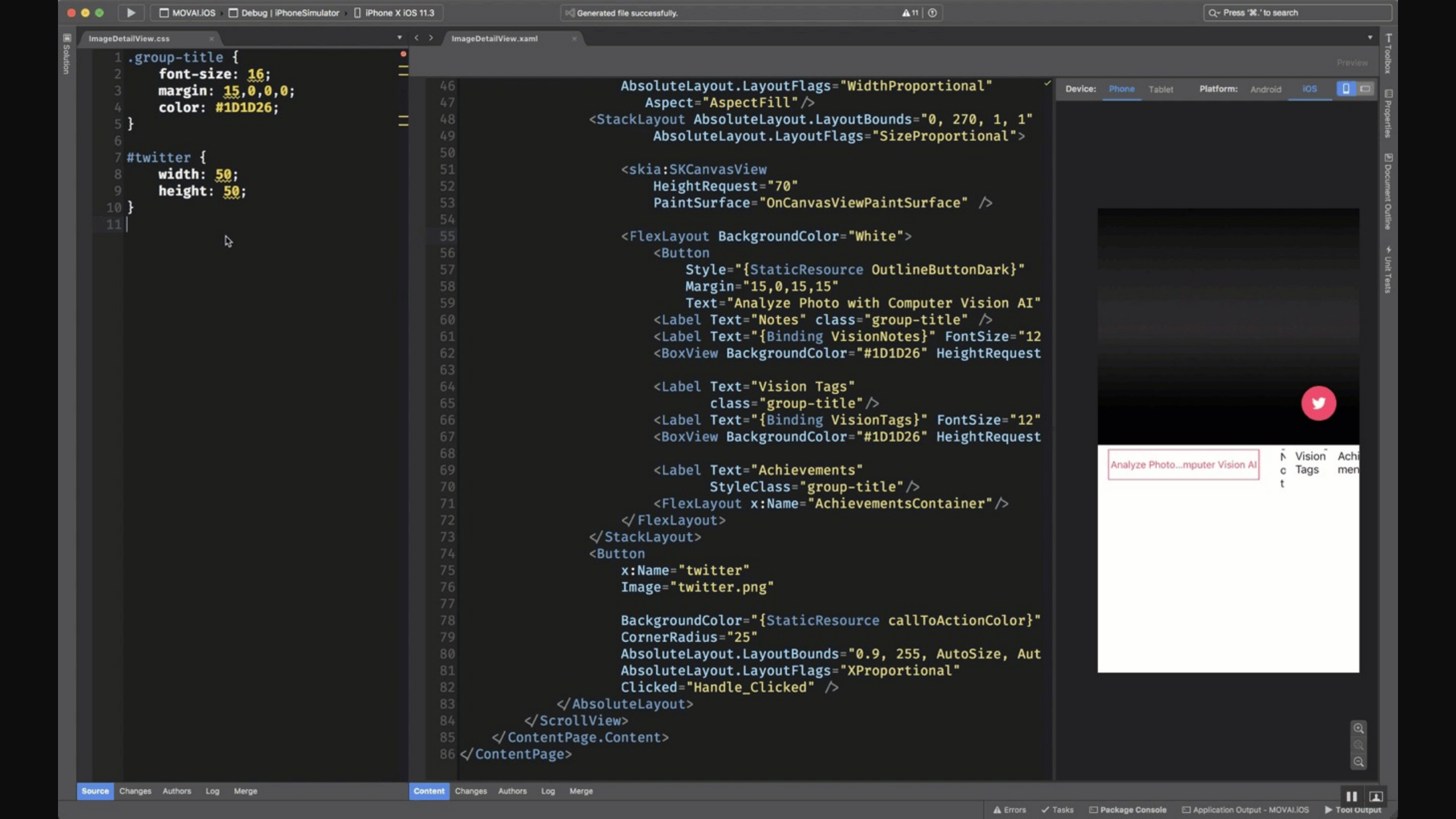
StyleSheets

- Styles expressed via Cascading Style Sheets (CSS)
- Parsed a runtime
- Does not replace XAML Styles
- Supports many properties
- But not all…
- Styles are applied one by one, in definition order.
- Styles defined on the item itself is always applied last.

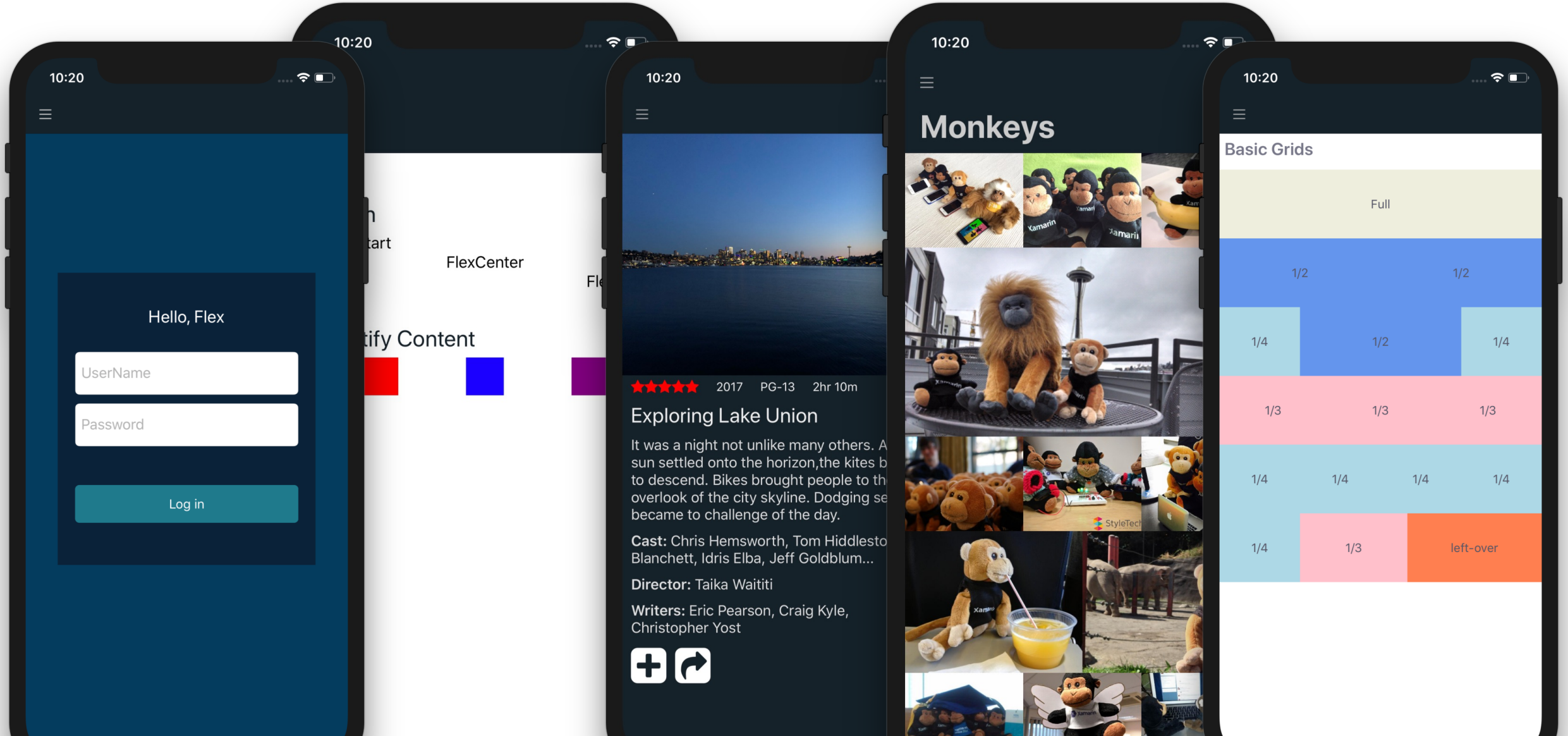
Demo

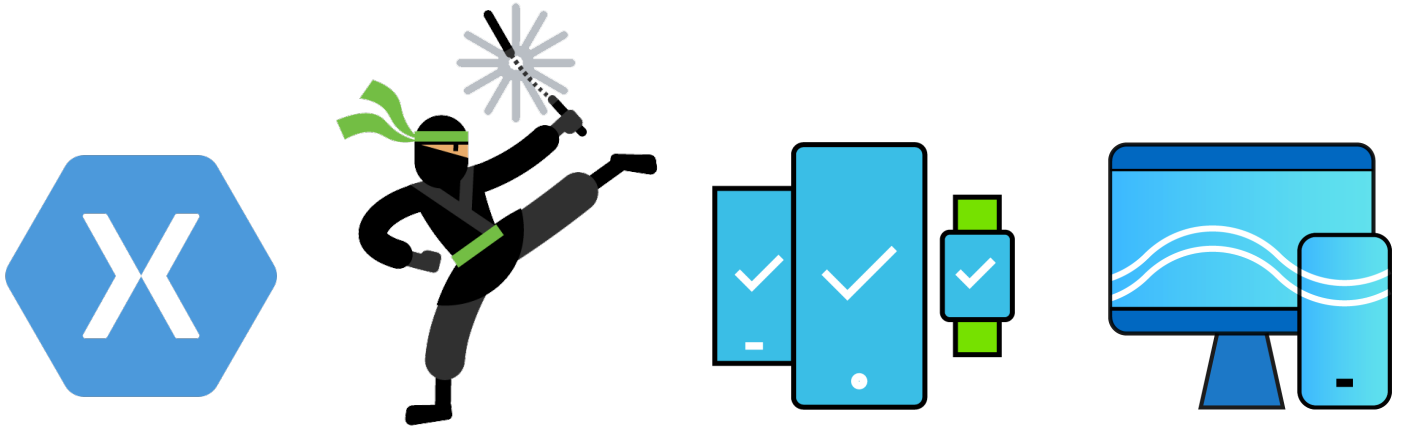



MOVAI.iOS › Debug | iPhoneSimulator › iPhone X iOS 11.3
Generated file successfully.
Press '⌘.' to search
Solution
AchievementsView.css
.container {
flex-direction: row;
flex-wrap: wrap;
justify-content: space-around;
align-items: flex-start;
}
.item {
padding: 0;
margin: 5,5;
height: 110;
width: 90;
}
AchievementsView.xaml
Preview
Device: Phone Tablet Platform: Android iOS
<ContentPage.Content>
<ScrollView
AbsoluteLayout.LayoutBounds="0,0,1,1"
AbsoluteLayout.LayoutFlags="SizeProportional">
<ScrollView.Margin>
<OnPlatform x:TypeArguments="Thickness" Default="15,0
<On Platform="Android" Value="15,50,15,0"/>
</OnPlatform>
</ScrollView.Margin>
<FlexLayout x:Name="AchievementsContainer"
>
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
<local:AchievementView class="item" />
</FlexLayout>
</ScrollView>
</ContentPage.Content>
</ContentPage>
Source Changes Authors Log Merge
Content Changes Authors Log Merge
Toolbox
Properties
Document Outline
Unit Tests
Errors Tasks Package Console Application Output - MOVAI.iOS Tool Output


MOVAI.iOS
Debug | iPhoneSimulator
iPhone X iOS 11.3
Generated file successfully.
Press '⌘.' to search
ImageDetailView.css
ImageDetailView.xaml
.group-title {
font-size: 16;
margin: 15,0,0,0;
color: #1D1D26;
}
#twitter {
width: 50;
height: 50;
}
AbsoluteLayout.LayoutFlags="WidthProportional"
Aspect="AspectFill" />
<StackLayout AbsoluteLayout.LayoutBounds="0, 270, 1, 1"
AbsoluteLayout.LayoutFlags="SizeProportional">
<skia:SKCanvasView
HeightRequest="70"
PaintSurface="OnCanvasViewPaintSurface" />
<FlexLayout BackgroundColor="White">
<Button
Style="{StaticResource OutlineButtonDark}"
Margin="15,0,15,15"
Text="Analyze Photo with Computer Vision AI"
<Label Text="Notes" class="group-title" />
<Label Text="{Binding VisionNotes}" FontSize="12
<BoxView BackgroundColor="#1D1D26" HeightRequest
<Label Text="Vision Tags"
class="group-title" />
<Label Text="{Binding VisionTags}" FontSize="12"
<BoxView BackgroundColor="#1D1D26" HeightRequest
<Label Text="Achievements"
StyleClass="group-title" />
<FlexLayout x:Name="AchievementsContainer" />
</FlexLayout>
</StackLayout>
<Button
x:Name="twitter"
Image="twitter.png"
BackgroundColor="{StaticResource callToActionColor}"
CornerRadius="25"
AbsoluteLayout.LayoutBounds="0.9, 255, AutoSize, Aut
AbsoluteLayout.LayoutFlags="XProportional"
Clicked="Handle_Clicked" />
</AbsoluteLayout>
</ScrollView>
</ContentPage.Content>
</ContentPage>
Preview
Device: Phone Tablet
Platform: Android iOS
Analyze Photo...mputer Vision AI
Vision Tags
Source Changes Authors Log Merge
Content Changes Authors Log Merge
Errors
Tasks
Package Console
Application Output - MOVAI.iOS
Tool Output

http://github.com/davidortinau/Flexibility





Modern X-Platform Xamarin.Forms UI

Sam Basu | @samidip

Telerik Developer Advocate | Progress

Resources

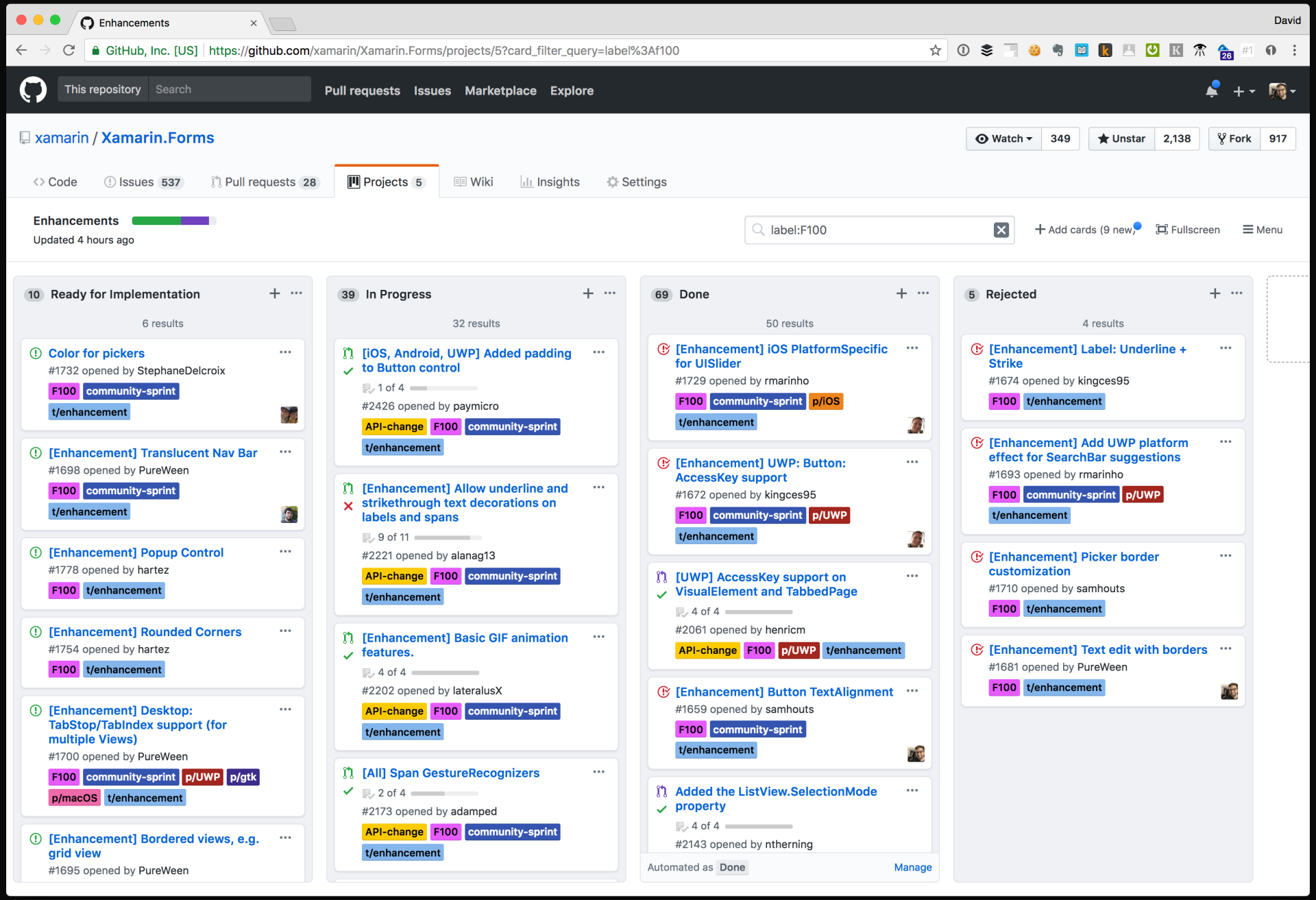
Microsoft Docs

https://docs.microsoft.com/en-us/xamarin/xamarin-forms/



https://university.xamarin.com/lightninglectures/

Future Roadmap


Enhancements
David
GitHub, Inc. [US] https://github.com/xamarin/Xamarin.Forms/projects/5?card_filter_query=label%3Af100
This repository Search
Pull requests Issues Marketplace Explore
xamarin / Xamarin.Forms
Watch 349 Unstar 2,138 Fork 917
Code Issues 537 Pull requests 28 Projects 5 Wiki Insights Settings
Enhancements
Updated 4 hours ago
label:F100
Add cards (9 new) Fullscreen Menu
10 Ready for Implementation
6 results
Color for pickers
#1732 opened by StephaneDelcroix
F100 community-sprint t/enhancement
[Enhancement] Translucent Nav Bar
#1698 opened by PureWeen
F100 community-sprint t/enhancement
[Enhancement] Popup Control
#1778 opened by hartez
F100 t/enhancement
[Enhancement] Rounded Corners
#1754 opened by hartez
F100 t/enhancement
[Enhancement] Desktop: TabStop/TabIndex support (for multiple Views)
#1700 opened by PureWeen
F100 community-sprint p/UWP p/gtk p/macOS t/enhancement
[Enhancement] Bordered views, e.g. grid view
#1695 opened by PureWeen
39 In Progress
32 results
[iOS, Android, UWP] Added padding to Button control
1 of 4
#2426 opened by paymicro
API-change F100 community-sprint t/enhancement
[Enhancement] Allow underline and strikethrough text decorations on labels and spans
9 of 11
#2221 opened by alanag13
API-change F100 community-sprint t/enhancement
[Enhancement] Basic GIF animation features.
4 of 4
#2202 opened by lateralusX
API-change F100 community-sprint t/enhancement
[All] Span GestureRecognizers
2 of 4
#2173 opened by adamped
API-change F100 community-sprint t/enhancement
69 Done
50 results
[Enhancement] iOS PlatformSpecific for UISlider
#1729 opened by rmarinho
F100 community-sprint p/iOS t/enhancement
[Enhancement] UWP: Button: AccessKey support
#1672 opened by kingces95
F100 community-sprint p/UWP t/enhancement
[UWP] AccessKey support on VisualElement and TabbedPage
4 of 4
#2061 opened by henricm
API-change F100 p/UWP t/enhancement
[Enhancement] Button TextAlignment
#1659 opened by samhouts
F100 community-sprint t/enhancement
Added the ListView.SelectionMode property
4 of 4
#2143 opened by ntherning
Automated as Done
Manage
5 Rejected
4 results
[Enhancement] Label: Underline + Strike
#1674 opened by kingces95
F100 t/enhancement
[Enhancement] Add UWP platform effect for SearchBar suggestions
#1693 opened by rmarinho
F100 community-sprint p/UWP t/enhancement
[Enhancement] Picker border customization
#1710 opened by samhouts
F100 t/enhancement
[Enhancement] Text edit with borders
#1681 opened by PureWeen
F100 t/enhancement

Roadmap

- vNext (3.1-pre1 Available Today!)
  - F100s including
    - Bindable Span
    - ListView SelectionMode
    - Drop Shadows
    - Hide ScrollView bars
    - Color on Slider, Switch
    - TitleView and Bottom Menu [In Progress]
- vNext2
  - F100s including
    - Rounded Corners
    - Rounded Image
    - Button Padding
  - Accessibility Improvements: Focus and Tab order
  - Gestures

Roadmap

- vNext3
  - x:Bind
  - LifeCycle Events

Research

Research

- Unified Design
- Performance
- Productivity

Proposals

MaterialShell
https://github.com/xamarin/Xamarin.Forms/issues/2415

Drawing
https://github.com/xamarin/Xamarin.Forms/issues/2452

Update to Xamarin.Forms 3.0 today!

Please Complete An Evaluation Form for every session you attend.

Your input is important!



#MSBuild

Microsoft

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